

MH1101 Tutorial 9 (Week 10)

Reference: Topic 5.1, 5.2 (Lecture Notes)

1. Use the Integral Test to determine whether the series is convergent or divergent. You need to explain why the corresponding function is decreasing on the domain under consideration.

(a) $\sum_{n=1}^{\infty} \frac{1}{(3n-1)^4}$

(b) $\sum_{n=1}^{\infty} \frac{n^3}{n^4 + 4}$

(c) $\sum_{n=1}^{\infty} ne^{-n^2}$

2. Determine whether the series converges or diverges.

(a) $\sum_{n=1}^{\infty} \frac{1}{n^3 + 8}$

(f) $\sum_{n=1}^{\infty} \frac{1}{n^{1+1/n}}$

(b) $\sum_{n=1}^{\infty} \frac{n-2}{n\sqrt{n}}$

(g) $\sum_{n=1}^{\infty} \frac{e^n + 1}{ne^n + 1}$

(c) $\sum_{n=1}^{\infty} \frac{9^n}{3 + 10^n}$

(h) $\sum_{n=1}^{\infty} \frac{\sqrt{1+n}}{2+n}$

(d) $\sum_{n=1}^{\infty} \frac{n^2 + n + 1}{n^4 + n^2}$

(i) $\sum_{n=1}^{\infty} \frac{n + 4^n}{n + 6^n}$

(e) $\sum_{n=1}^{\infty} \sin\left(\frac{1}{n}\right)$

(j) $\sum_{n=1}^{\infty} \frac{\tan^{-1} n}{n^{1.2}}$

3. Find the values of p for which the series is convergent:

$$\sum_{n=1}^{\infty} n(1+n^2)^p.$$

4. (The Limit Comparison Test) Suppose $\sum a_n$ and $\sum b_n$ are series with positive terms. Prove that if $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \infty$ and $\sum b_n$ diverges, then $\sum a_n$ diverges.

5. Suppose $b_n > 0$ and $\sum_{n=1}^{\infty} b_n$ is convergent. Prove that the following series is also convergent.

(i) $\sum_{n=1}^{\infty} \ln(1 + b_n)$

(ii) $\sum_{n=1}^{\infty} \sin(b_n)$

Hint: Use the Limit Comparison Test, and the l'Hospital's Rule.

Answer Keys. 1. (a) converges (b) diverges (c) converges 2. (a) converges (b) diverges (c) converges (d) converges (e) diverges (f) diverges (g) diverges (h) diverges (i) converges (j) converges 3. $p < -1$